# Notes: Bloom, Draca, and Van Reenen (RES, 2016)

**Trade Induced Technical Change? The Impact of Chinese Imports on Innovation, IT and Productivity**

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# 1 Big picture

- **Question.** Does import competition from a low-wage country like China simply destroy jobs in richer countries, or can it also *induce technical change* inside firms and reallocate activity toward more technologically advanced producers?
- **Setting.** The paper studies manufacturing firms and plants in twelve European countries over the late 1990s and 2000s, the period when Chinese imports rose sharply and China entered the WTO in 2001.
- **Core empirical object.** The main treatment is the growth of Chinese import penetration in a country–industry cell, mapped to firms through their four-digit industry exposures.
- **Main finding.** Higher Chinese import competition raises patenting, increases IT intensity, and improves TFP within surviving firms. At the same time, it reduces employment and survival among low-tech firms, so trade also upgrades technology through between-firm reallocation.
- **Magnitude.** The paper's baseline calculations imply that Chinese imports account for about 14% of European technology upgrading between 2000 and 2007 in patents, IT, and TFP. The implied contribution is even larger once offshoring and IV estimates are incorporated.
- **Mechanism / intuition.** The paper argues that trade with China makes low-tech production less attractive in Europe. Firms respond by innovating, adopting new technologies, improving management, and shifting resources toward more advanced activities. Low-tech firms shrink and exit, while more advanced firms are relatively protected.
- **Why this matters.** A large trade literature had emphasized selection and reallocation across firms. This paper shows that trade also changes technology *within* firms, using direct measures like patents and computers per worker instead of only indirect productivity proxies.
- **Additional punchline.** The technology effects are specific to low-wage import competition. Imports from developed countries do not have significant effects on innovation in the same regressions.

# 2 Data and measurement (key objects)

## 2.1 Core firm and plant data

- **Base source.** The paper starts from Bureau van Dijk's *Amadeus*, which contains close to the population of public and private firms in twelve European countries: Austria, Denmark, Finland, France, Germany, Ireland, Italy, Norway, Spain, Sweden, Switzerland, and the UK.
- **Industry mapping.** Firms have primary and secondary four-digit industry codes. The paper maps trade exposure to firms using a weighted average of Chinese imports across the industries in which the firm operates.
- **Unit of observation.** The patent and TFP analyses are at the firm level; the IT analysis is at the establishment / plant level because the Harte Hanks data are establishment based.

## 2.2 Patents

- **Source.** The paper matches Amadeus to the population of patent applications at the European Patent Office.
- **Timing.** Patents are dated by application year so they proxy for invention timing rather than grant timing.
- **Matching.** Patent assignee names are matched to Amadeus firms using names and location, following Belenzon and Berkovitz (2010).
- **Samples.** The authors analyze both a baseline sample of *patenters* (firms that filed at least one EPO patent since 1978) and a broader sample in which unmatched firms are assigned zero patents.
- **Quality control.** Because patent values are heterogeneous, some specifications examine citations per patent to test whether the results reflect true innovation rather than a defensive "lawyer effect."

## 2.3 Productivity and exit

- **Accounting data.** Amadeus provides employment, capital, materials, wage bills, and sales.
- **TFP sample.** The TFP analysis uses France, Italy, Spain, and Sweden because these countries have near-population coverage and include materials, which are needed for three-factor production-function estimation.
- **Construction.** The core TFP measure uses the De Loecker (2011) adaptation of Olley and Pakes (1996), estimated separately by two-digit industry and allowing for trade, imperfect competition, and multi-product firms.
- **Exit.** A firm is coded as exiting when its Amadeus status indicates "bankrupt," "liquidated," or "dormant." Mergers and takeovers are not counted as exit.

## 2.4 Information technology

- **Source.** The paper uses Harte Hanks data on roughly 160,000 European establishments, restricting to the same twelve countries as the patent analysis.
- **Main measure.** The central IT variable is computers per worker (PCs plus laptops). The authors prefer this because it is a physical quantity measure and avoids cross-country price-deflator problems.
- **Why Harte Hanks is useful.** The data are collected for commercial use by large IT vendors, which helps discipline data quality.
- **Coverage.** Harte Hanks surveys all firms with more than 100 employees, so the IT sample is tilted toward larger establishments, though the paper reports little evidence this induces major bias.
- **Other IT margins.** Robustness checks use ERP, database software, and groupware adoption.

### 2.5 Trade exposure

- **Trade source.** Bilateral import data come from UN Comtrade at the six-digit product level, aggregated to four-digit US SIC industries using the Pierce–Schott concordance.
- **Main measure.** The key exposure variable is

$$\text{IMPCH}_{jkt} = \frac{M_{jkt}^{\text{China}}}{M_{jkt}^{\text{World}}},$$

  the share of imports in industry $j$ and country $k$ that originate from China.
- **Interpretation.** A rise in IMPCH means China is taking a larger share of import demand in that country–industry, which the paper treats as a proxy for tougher low-wage competition.
- **Alternatives.** The paper also normalizes Chinese imports by domestic production or by apparent consumption and obtains similar results.

### 2.6 Quota-based instrument

- **Institutional source.** The main IV exploits China's WTO accession in 2001 and the abolition of textile and apparel quotas under the Agreement on Textiles and Clothing / Multi-Fiber Agreement.
- **Instrument definition.** For each four-digit industry, the instrument is the value-weighted share of six-digit products that were covered by a quota against China in 2000.
- **Why it matters.** When quotas were abolished, Chinese imports in affected product groups rose by about 240% on average.
- **Cross-industry variation.** The pre-2000 quota coverage varied a lot: for example, 77% of cotton fabric products were covered but only 2% of wool fabric products; 100% of women's dresses were covered but only 5% of men's trousers.
- **Exclusion idea.** The quotas were old political bargains, largely built up from the 1950s onward. The paper argues that this historical protection pattern is plausibly unrelated to differential technology shocks in the early 2000s.

### 2.7 Descriptive statistics

- **China's rise.** China's share of all imports to the US and the twelve European countries rose from about 5.7% in 2000 to 12.4% in 2007.
- **Sectoral pattern.** The rise was especially strong in industries like toys, furniture, and footwear.
- **Patent sample.** In the patenting sample, the mean number of patents per year is about 1 and median employment is about 100.
- **TFP sample.** Median employment is about 30, reflecting the fact that this sample is wider and includes many firms that do not patent.
- **IT sample.** Median employment is about 140 and mean IT intensity is about 0.58 computers per worker.

## 3 Models and empirical specifications (all equations + interpretation)

### 3.1 Baseline firm-level technology equation

The paper begins from a firm-level technology equation:

$$\ln TECH_{ijkt} = \alpha \,\mathrm{IMPCH}_{jkt-\ell} + \eta_i + f_{kt} + \varepsilon_{ijkt}. \tag{1}$$

- **Dependent variable.** $TECH_{ijkt}$ is measured alternately by patents, computers per worker, or TFP.
- **Regressor of interest.** IMPCH is Chinese imports as a share of total imports in country–industry $(j, k)$.
- **Fixed effects.** $\eta_i$ is a firm fixed effect and $f_{kt}$ is a country-by-year effect.
- **Prediction.** If Chinese competition induces technical change, then $\alpha > 0$.
- **Lag structure.** The notation allows for dynamic responses through the lag $\ell$, though the baseline estimates set $\ell = 0$ for consistency.

### 3.2 Long-difference estimating equation

To eliminate firm fixed effects, the main estimating equation is in long differences:

$$\Delta \ln TECH_{ijkt} = \alpha \,\Delta\mathrm{IMPCH}_{jkt} + \Delta f_{kt} + \Delta\varepsilon_{ijkt}. \tag{2}$$

- **Implementation.** The authors usually use five-year differences, often overlapping, such as 2005–2000 and 2004–1999.
- **Interpretation.** A positive coefficient means firms in industries where Chinese import share rises faster also upgrade technology faster.
- **Inference.** Standard errors are clustered at the country–industry pair level in most OLS specifications.
- **Important reading point.** Because employment tends to fall in exposed firms, the paper emphasizes that patent *levels* rise, not just patents per worker.

### 3.3 Quota-based first stage and reduced form

To address endogeneity, the paper instruments trade exposure using quota abolition:

$$\Delta\mathrm{IMPCH}_{jkt} = -\phi \,\Delta\mathrm{QUOTA}_{jkt} + \Delta f^{Q}_{kt} + \Delta\varepsilon^{Q}_{ijkt}. \tag{3}$$

For the main 2000–2005 difference, because quotas are eliminated by 2005, this becomes

$$\Delta\mathrm{IMPCH}_{jkt} = \phi \,\mathrm{QUOTA}_{jk,2000} + \Delta f^{Q}_{kt} + \Delta\varepsilon^{Q}_{ijkt}. \tag{4}$$

The associated reduced form for technology is

$$\Delta \ln TECH_{ijkt} = \pi \, \mathrm{QUOTA}_{jk,2000} + \Delta \varsigma_{kt} + \Delta e_{ijkt}. \tag{5}$$

- **Relevance.** Industries with tougher initial quotas experience faster subsequent Chinese import growth when quotas are removed.
- **Exclusion restriction.** Pre-2000 quota toughness should affect post-2000 innovation only through Chinese import competition, not through unrelated technology shocks.

### 3.4 Employment-growth equation and reallocation logic

The paper then studies between-firm reallocation using employment growth:

$$\begin{aligned} \Delta \ln N_{ijkt} = \alpha^N \Delta \mathrm{IMPCH}_{jkt} + \gamma^N \left( TECH_{ijkt-5} \times \Delta \mathrm{IMPCH}_{jkt} \right) \\ + \delta^N TECH_{ijkt-5} + \Delta f^N_{kt} + \Delta \varepsilon^N_{ijkt}. \end{aligned} \tag{6}$$

- **Dependent variable.** $N_{ijkt}$ is employment.
- **Expected signs.** The paper expects $\alpha^N < 0$: Chinese imports reduce jobs on average. It expects $\gamma^N > 0$: firms with better initial technology are more shielded.
- **Economic meaning.** This is the paper's main selection / reallocation equation. It asks whether low-tech firms are hit disproportionately hard by trade competition.

### 3.5 Survival equation

The corresponding survival specification is

$$\begin{aligned} SURVIVAL_{ijkt} = S_{ijkt} = \alpha^S \Delta \mathrm{IMPCH}_{jkt} + \gamma^S \left( TECH_{ijkt-5} \times \Delta \mathrm{IMPCH}_{jkt} \right) \\ + \delta^S TECH_{ijkt-5} + \Delta f^S_{kt} + \Delta \varepsilon^S_{ijkt}. \end{aligned} \tag{7}$$

- **Dependent variable.** $S_{ijkt} = 1$ if a firm / establishment alive in the base year survives for the next five years.
- **Expected signs.** If Chinese competition raises exit risk, then $\alpha^S < 0$. If high-tech firms are more protected, then $\gamma^S > 0$.
- **Implementation.** When this equation is estimated with IV, both $\Delta$IMPCH and its interaction with lagged technology are instrumented.

### 3.6 Industry-level technology equation

Because entry cannot be handled cleanly at the firm level, the paper also estimates an industry-level analogue:

$$\Delta TECH_{jkt} = \alpha^{IND} \Delta \mathrm{IMPCH}_{jkt} + \Delta f^{IND}_{kt} + \Delta \varepsilon^{IND}_{jkt}. \tag{8}$$

- **Purpose.** This specification captures within-firm effects, between-firm reallocation, and entry effects together.
- **Interpretation.** Comparing industry-level estimates with firm-level estimates helps the authors gauge the importance of selection and entry relative to pure within-firm upgrading.

### 3.7 Aggregate decomposition used for magnitudes

To quantify the economy-wide importance of China, the paper decomposes aggregate technology growth:

$$\begin{aligned}\Delta P_t = &\sum_{i=1}^{N} s_{i0}(p_{ijt} - p_{ij0}) + \sum_{i=1}^{N}(s_{it} - s_{i0})p_{ij0} + \sum_{i=1}^{N}(s_{it} - s_{i0})(p_{ijt} - p_{ij0}) \\ &- \sum_{i \in exit} s_{it}^{exit}(p_{ij0}^{exit} - \bar{p}_{j0}) + \sum_{i \in entrant} s_{it}^{entrant}(p_{ijt}^{entrant} - \bar{p}_{jt}).\end{aligned} \tag{9}$$

- **First term.** Within-firm technology upgrading.
- **Second and third terms.** Between-firm reallocation and the cross term.
- **Fourth term.** Exit contribution.
- **Fifth term.** Entry contribution.
- **Why this matters.** The decomposition is what lets the paper translate regression coefficients into statements like "China explains 13.9% of aggregate patenting growth."

## 4 Identification and instruments (what makes the estimates credible)

### 4.1 Baseline identification idea

- The most basic identification comes from comparing firms exposed to larger versus smaller increases in Chinese import penetration within the same country-year or, in tougher specifications, within the same narrow industry-year.
- The concern is obvious: technology shocks can themselves affect trade exposure. For example, an unobserved domestic technology improvement could reduce Chinese market share, generating downward bias in OLS.
- The paper therefore treats the direction of OLS bias as an empirical question and turns to instruments.

### 4.2 Quota-removal IV

- **Instrument.** The 2000 level of quota coverage against China in textiles and clothing.
- **Relevance.** When quotas are abolished after WTO accession, industries with tighter pre-2000 quotas experience larger import surges from China.
- **First-stage strength.** Table 2 reports first-stage F-statistics of 24.1 for patents, 21.4 for IT, and 11.5 for TFP in the textiles / clothing subsample.
- **Exclusion restriction.** The authors argue that quota coverage reflects historic political bargaining and not early-2000s technology shocks.

### 4.3 Why the exclusion restriction looks plausible in the paper

- The industries with high quota coverage do *not* exhibit differential pre-2000 trends in patents, TFP, labour productivity, capital intensity, materials intensity, average wages, employment, or capital.
- The reduced-form technology results remain when the authors add firm-specific trends, effectively estimating trend-adjusted difference-in-differences equations.
- The paper also checks for anticipation effects and finds no evidence that firms in heavily protected sectors were already slowing innovation before WTO accession.
- The political reintroduction of some temporary quotas after 2005 also supports the idea that full liberalization was not perfectly anticipated.

### 4.4 Initial-conditions IV

- **Alternative instrument.** Lagged Chinese import share in an industry across Europe and the US, interacted with aggregate Chinese import growth.
- **Intuition.** Industries where China already had a revealed comparative advantage in the late 1990s are the industries where Chinese import penetration subsequently rises most when China expands globally.
- **Advantage.** Unlike the quota IV, this strategy can be used outside textiles and clothing and thus covers the full sample.
- **Empirical message.** The initial-conditions IV delivers results qualitatively similar to the quota IV and often larger than OLS.

### 4.5 Alternative channels and interpretation

- **Offshoring.** The paper explicitly measures Chinese offshoring following Feenstra and Hanson. Offshoring helps explain IT and TFP gains, but it does not eliminate the positive effect of final-goods competition on technology.
- **Product switching.** Firms do shift product mix in response to trade, but the increase in the absolute volume of patents suggests the story is not only compositional. Firms are creating ideas that are new to the world.
- **Developed-country imports.** Imports from advanced countries do not have the same positive effect on innovation, which supports the paper's emphasis on low-wage-country competition rather than generic import pressure.

## 5 Tables and figures

### 5.1 Figure 1: Share of all imports in the EU and US from China and all low-wage countries

**Read:**

- The figure shows a dramatic rise in import exposure from China, especially after the mid-1990s. China's share rises from a very low base in the early 1980s to more than 10% by 2007.
- The broader low-wage-country share also rises, but the increase is increasingly dominated by China. That is why the paper treats China as the key low-wage trade shock.
- This figure is mostly descriptive, but it is the visual motivation for the entire paper: the period under study contains a quantitatively major and plausibly exogenous shift in trade exposure.

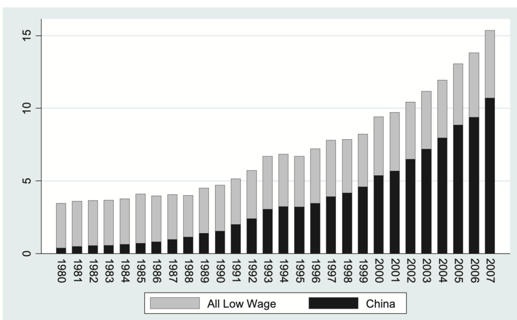


FIGURE 1
Share of all imports in the EU and US from China and all low-wage countries

*Notes*: Calculated using UN Comtrade data. Low-wage countries list taken from Bernard *et al.* (2006) and are defined as countries with less than 5% GDP/capita relative to the US 1972–2001.

TABLE 1
*Technical change within incumbent firms and plants*

| Panel A: Baseline results | | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Dependent variable | $\Delta$ln(PATENTS) | $\Delta$ln(IT/$N$) | $\Delta$TFP |
| Estimation method | Five-year differences | Five-year differences | Five-year differences |
| Change in Chinese imports | 0.321*** | 0.361** | 0.257*** |
| $\Delta IMP_{jk}^{CH}$ | (0.102) | (0.076) | (0.072) |
| Sample period | 2005–1996 | 2007–2000 | 2005–1995 |
| Number of units | 8,480 | 22,957 | 89,369 |
| Number of country by industry clusters | 1,578 | 2,816 | 1,210 |
| Observations | 30,277 | 37,500 | 292,167 |
| **Panel B: Include industry trends** | | | |
| Dependent variable | $\Delta$ln(PATENTS) | $\Delta$ln(IT/$N$) | $\Delta$TFP |
| Change in Chinese imports | 0.191* | 0.170** | 0.128** |
| $\Delta IMP_{jk}^{CH}$ | (0.102) | (0.082) | (0.053) |
| Number of units | 8,480 | 22,957 | 89,369 |
| Number of country by industry clusters | 1,578 | 2,816 | 1,210 |
| Observations | 30,277 | 37,500 | 292,167 |
| **Panel C: Normalize imports by domestic production** | | | |
| Dependent variable | $\Delta$ln(PATENTS) | $\Delta$ln(IT/$N$) | $\Delta$TFP |
| Change in Chinese imports | 0.142** | 0.129*** | 0.065*** |
| $\Delta IMP_{jk}^{CH}$ | (0.048) | (0.028) | (0.020) |
| Number of units | 8,474 | 20,106 | 89,369 |
| Number of country by industry clusters | 1,575 | 2,480 | 1,210 |
| Observations | 30,221 | 31,820 | 292,167 |
| **Panel D: Offshoring** | | | |
| Dependent variable | $\Delta$ln(PATENTS) | $\Delta$ln(IT/$N$) | $\Delta$TFP |
| Change in Chinese imports | 0.313*** | 0.279*** | 0.188*** |
| $\Delta IMP_{jk}^{CH}$ | (0.100) | (0.080) | (0.082) |
| Change Chinese imports | 0.174 | 1.685*** | 1.396*** |
| in source industries $\Delta$OFFSHORE | (0.822) | (0.517) | (0.504) |
| Number of units | 8,480 | 22,957 | 89,369 |
| Number of country by industry clusters | 1,578 | 2,816 | 1,210 |
| Observations | 30,277 | 37,500 | 292,167 |

*Notes*: *** denotes 1% significance; ** denotes 5% significance; and * denotes 10% significance. Sample period is the same in all panels, i.e., 2005–1996 for Column (1); 2007–2000 for Column (2), and 2005–1995 for Column (3). Estimation is by OLS with standard errors clustered by country by four-digit industry pair in parentheses. All changes are in five-year differences, e.g., $\Delta IMP_{jk}^{CH}$ represents the five-year difference in Chinese imports as a fraction of total imports in a four-digit industry by country pair. All columns include a full set of country by year dummies. $\Delta$ln(PATENTS) is the change in ln(1+PAT), PAT = count of patents. IT/$N$ is the number of computers per worker. TFP is estimated using the De Loecker (2011) version of the Olley–Pakes (1996) method separately for each industry (see online Appendix C). Panel B includes three-digit industry trends. Panel C normalizes Chinese imports on domestic production (instead of total imports as in other columns). Panel D includes a measure of offshoring defined as in Feenstra and Hanson (1999) except it is for Chinese imports only, not all low-wage country imports (see online Appendix A). The twelve countries include Austria, Denmark, Finland, France, Germany, Ireland, Italy, Norway, Spain, Sweden, Switzerland, and the UK for all columns except (3) which only includes France, Italy, Spain, and Sweden (the countries where we have good data on intermediate inputs). Dummies for establishment type (Divisional HQ, Divisional Branch, Enterprise HQ or a Standalone Branch) are included in Column (2). Units are firms in Columns (1) and (3) and plants in Column (2).

## 5.2 Table 1: Technical change within incumbent firms and plants

**Read:**

- Panel A is the core within-firm result: the coefficient on Chinese imports is positive and significant for patents (0.321), computers per worker (0.361), and TFP (0.257).
- Economically, a higher growth rate of Chinese import penetration is associated with more innovation and faster technology adoption inside surviving firms.
- Panel B adds industry trends. Coefficients shrink but remain positive and significant, which is important because it means the result is not driven by broad technology trends within industries.
- Panel C normalizes imports by domestic production rather than total imports, and the conclusions are unchanged.
- Panel D adds a Chinese offshoring measure. Offshoring matters for IT and TFP, but the direct effect of Chinese import competition in final goods remains strongly positive.

## 5.3 Table 2: Within-firm IV results using quota removal in textiles and clothing

**Read:**

- The table narrows to clothing and textiles, where the quota instrument is relevant.
- OLS is already large: 1.160 for patents, 1.284 for IT, and 0.902 for TFP.
- IV coefficients are even larger: 1.864 for patents, 1.851 for IT, and 1.629 for TFP. The first-stage F-statistics are 24.1, 21.4, and 11.5.
- The main reading is that OLS does *not* appear upward biased. If anything, the paper's baseline OLS estimates may understate the effect of Chinese competition on technology.

TABLE 2

*Within firm results—using changes in quotas as an IV for chinese imports (clothing and textile industries only)*

| | (1) | (2) Patenting activity | (3) | (4) | (5) Information technology | (6) | (7) | (8) Total factor productivity | (9) |
|---|---|---|---|---|---|---|---|---|---|
| Dependent variable | $\Delta\ln$(PATENTS) | $\Delta IMP^{CH}$ | $\Delta\ln$(PATENTS) | $\Delta\ln(IT/N)$ | $\Delta IMP^{CH}$ | $\Delta\ln(IT/N)$ | $\Delta$TFP | $\Delta IMP^{CH}$ | $\Delta$TFP |
| Method | OLS | First stage | IV | OLS | First stage | IV | OLS | First stage | IV |
| Change Chinese imports | 1.160*** | | 1.864* | 1.284*** | | 1.851*** | 0.902*** | | 1.629** |
| | (0.377) | | (1.001) | (0.172) | | (0.397) | (0.118) | | (0.326) |
| Quotas removal | | 0.108*** | | | 0.088*** | | | 0.107*** | |
| QUOTA | | (0.022) | | | (0.019) | | | (0.032) | |
| $F$-statistic | | 24.1 | | | 21.4 | | | 11.5 | |
| Sample period | 2005–1999 | 2005–1999 | 2005–1999 | 2005–2000 | 2005–2000 | 2005–2000 | 2005–1999 | 2005–1999 | 2005–1999 |
| Number of units | 1,866 | 1,866 | 1,866 | 2,891 | 2,891 | 2,891 | 12,247 | 12,247 | 12,247 |
| Number industry clusters | 149 | 149 | 149 | 83 | 83 | 83 | 177 | 177 | 177 |
| Observations | 3,443 | 3,443 | 3,443 | 2,891 | 2,891 | 2,891 | 20,625 | 20,625 | 20,625 |

*Notes*: *** denotes 1% significance; ** denotes 5% significance; and * denotes 10% significance. In all panels we use the same specifications as Table 1 Columns (1), (2), and (3) but estimate by instrumental variables (IV). In Panel A the IV is "Quota removal" is based on EU SIGL data and defined as the (value weighted) proportion of HS6 products in the four-digit industry that were covered by a quota restriction on China in 2000 (prior to China's WTO accession) that were planned to be removed by 2005 (see online Appendix C for details). The number of units is the number of firms in all columns except the IT specification where it is the number of plants. All columns include country by year effects. Sample is firms in the clothing and textile. Standard errors for all regressions are clustered by four-digit industry in parentheses (the quota IV is defined at the SIC4 industry level and does not vary across countries like the Chinese import share, which is why we take the more conservative approach to clustering compared with Table 1).

TABLE 3

*Within firm effects—including firm-specific trends with quotas; textile and clothing industry*

| | Patenting | | | | Total factor productivity | | | |
|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Dependent variable | $\Delta\ln$(PATENTS) | $\Delta\ln$(PATENTS) | $\Delta\ln$(PATENTS) | $\Delta\ln$(PATENTS) | $\Delta$TFP | $\Delta$TFP | $\Delta$TFP | $\Delta$TFP |
| Quotas removal | 0.197** | 0.216** | | | 0.152*** | 0.178*** | | |
| *I(year >2000) | (0.095) | (0.105) | | | (0.037) | (0.037) | | |
| Quotas removal | | | 0.066** | 0.075** | | | 0.028** | 0.033* |
| * # years after 2000 | | | (0.029) | (0.033) | | | (0.014) | (0.017) |
| Firm-specific trends? | No | Yes | No | Yes | No | Yes | No | Yes |
| Sample period | 2005–1992 | 2005–1992 | 2005–1992 | 2005–1995 | 2005–1995 | 2005–1995 | 2005–1995 | 2005–1995 |
| Number of firms | 2,435 | 2,435 | 2,435 | 2,435 | 16,495 | 16,495 | 16,495 | 16,495 |
| Number industry clusters | 159 | 159 | 159 | 159 | 187 | 187 | 187 | 187 |
| Observations | 14,768 | 14,768 | 14,768 | 14,768 | 55,791 | 55,791 | 55,791 | 55,791 |

*Notes*: *** denotes 1% significance; ** denotes 5% significance; and * denotes 10% significance. These are the equivalent of the reduced forms underlying Table 2. We use a longer sample period than Table 2 in order to include trends. "Quota removal" (QUOTA) is based on EU SIGL data and defined as the (value weighted) proportion of HS6 products in the four-digit industry that were covered by a quota restriction on China in 2000 (prior to China's WTO accession) that were planned to be removed by 2005 (see online Appendix C for details). I(year >2000) is an indicator variable = 1 if observation is after 200 (i.e., after China's WTO accession). "# years after 2000" is the number of years after 2000 and zero in 2000 and before (i.e., "# years after 2000" = 1 in 2001, = 2 in 2002, etc.). All estimates are in five-year differences as usual, so we control for firm-specific trends by including a firm dummy in Columns (2), (4), (6), and (8). All columns include country by year effects. Sample is firms in the clothing and textile industry. Standard errors for all regressions are clustered by four-digit industry in parentheses (the quota IV does not vary across within industry across countries like the Chinese import share, which is why we take the more conservative approach compared with Table 1).

## 5.4 Table 3: Reduced-form quota results with firm-specific trends

**Read:**

- This is the paper's toughest trend test. It interacts quota exposure either with a post-2000 dummy or with the number of years after 2000.
- For patenting, the post-2000 reduced form is 0.197 without firm trends and 0.216 with firm trends. For the linear time-since-2000 version, the coefficients are 0.066 and 0.075.
- For TFP, the corresponding coefficients are also positive: 0.152 and 0.178 in the post-2000 specification, and 0.028 and 0.033 in the time-since-2000 specification.
- The point is simple: industries more exposed to quota abolition innovate more after WTO accession even after controlling for firm-specific trends.

## 5.5 Table 4: Between-firm effects on employment and survival

**Read:**

- Panel A shows that Chinese imports reduce employment growth. In the patent sample, a higher import shock is associated with a strong fall in jobs, with a baseline coefficient of −0.361.
- The interaction between imports and lagged technology is positive: 1.435 for patents, 0.385 for IT, and 0.795 for TFP. High-tech firms are more insulated, while low-tech firms are hit hardest.
- Panel B shows a similar pattern for survival. Chinese imports reduce survival probabilities, and the interaction with lagged technology is again positive in the patent and TFP specifications.
- This table is the cleanest evidence for the paper's *reallocation* channel: Chinese trade not only induces within-firm upgrading, it also shifts employment away from technologically weaker firms.

TABLE 4

*Between firm effects—employment and survival*

| Panel A: Employment $\Delta \ln N$ | | | | | | |
|---|---|---|---|---|---|---|
| Dependent variable: Employment growth, $\Delta \ln N$ | (1) | (2) | (3) | (4) | (5) | (6) |
| Technology variable (TECH) | PATENTS | PATENTS | IT | IT | TFP | TFP |
| Change in Chinese imports | −0.361*** | −0.434*** | −0.203** | −0.379*** | −0.377*** | −0.377*** |
| $\Delta \mathrm{IMP}_{jk}^{CH}$ | (0.134) | (0.136) | (0.086) | (0.105) | (0.094) | (0.096) |
| Change in Chinese imports * technology at $t-5$ | | 1.435** | | 0.385** | | 0.795** |
| $\Delta \mathrm{IMP}_{jk}^{CH} * TECH_{t-5}$ | | (0.649) | | (0.157) | | (0.307) |
| Technology at $t-5$ | 0.389*** | 0.348*** | 0.241*** | 0.230*** | 0.152*** | 0.136*** |
| $\mathrm{TECH}_{t-5}$ | (0.043) | (0.049) | (0.010) | (0.010) | (0.012) | (0.012) |
| Number of units | 6,335 | 6,335 | 22,957 | 22,957 | 89,369 | 89,369 |
| Countries by industry clusters | 1,376 | 1,376 | 2,816 | 2,816 | 1,210 | 1,210 |
| Observations | 19,844 | 19,844 | 37,500 | 37,500 | 292,167 | 292,167 |
| **Panel B: Survival** | | | | | | |
| Dependent Variable: SURVIVAL | (1) | (2) | (3) | (4) | (5) | (6) |
| Technology variable (TECH) | PATENTS | PATENTS | IT | IT | TFP | TFP |
| Change in Chinese imports | −0.057 | −0.078 | −0.118** | −0.182** | −0.207*** | −0.208*** |
| $\Delta \mathrm{IMP}_{jk}^{CH}$ | (0.046) | (0.049) | (0.047) | (0.072) | (0.051) | (0.050) |
| Change in Chinese imports * technology at $t-5$ | | 0.260** | | 0.137 | | 0.110* |
| $\Delta \mathrm{IMP}_{jk}^{CH} * \mathrm{TECH}_{t-5}$ | | (0.122) | | (0.112) | | (0.059) |
| Technology at $t-5$ | 0.003 | −0.005 | 0.001 | −0.002 | −0.001 | −0.004 |
| $\mathrm{TECH}_{t-5}$ | (0.007) | (0.009) | (0.005) | (0.006) | (0.003) | (0.003) |
| Survival rate for sample (mean) | 0.977 | 0.977 | 0.886 | 0.886 | 0.927 | 0.927 |
| Number of country by industry clusters | 1,647 | 1,647 | 2,863 | 2,863 | 1,242 | 1,242 |
| Observations (and number of units) | 7,928 | 7,928 | 28,624 | 28,624 | 60,883 | 60,883 |

*Notes*: *** denotes 1% significance; ** denotes 5% significance; and * denotes 10% significance. Estimation by OLS with standard errors (clustered by country by four-digit industry pair) in parentheses. $\Delta \mathrm{IMP}^{CH}$ is the five-year difference in Chinese imports as a fraction of total imports in a four-digit industry by country pair. In Columns (1) and (2) TECH is ln[(1+ the firm's patent stock)/employment]; in Columns (3) and (4) TECH is computers per employee, and in Columns (5) and (6) it is TFP. Twelve countries in all columns except Columns (5) and (6) which is four countries. Number of units is the number of firms in all columns except IT where it is the number of plants. All columns include country by year effects. In Panel A, the dependent variable is the five-year difference of ln(employment). The sample period is 2005–1996 for patents, 2007–2000 for IT, and 2005–1995 for TFP. In Panel B, the sample period is the 2000–2005 cross-section. The dependent variable is SURVIVAL which refers to whether an establishment in Columns (3) or (4) or firm (in all other columns) that was alive in 2000 was still alive in 2005. Specifically, we classify an establishment as having exited if it drops out of the panel and does not appear for four successive years in Columns (3) and (4). In the other columns SURVIVAL it is based on Amadeus company status (see online Appendix B) where exit is defined on the basis of whether a firm that was active in 2000 is recorded as either "bankrupt", "liquidated", or "dormant" in the Company Status variable provided by BVD in 2005 and beyond.

TABLE 5

*Between firm effects—using quota removal as an IV for Chinese imports*

| Panel A: Employment | | | | | | |
|---|---|---|---|---|---|---|
| Dependent variable: Employment growth | (1) | (2) | (3) | (4) | (5) | (6) |
| Technology variable (TECH) | PATENTS | PATENTS | IT | IT | TFP | TFP |
| Estimation technique | OLS | IV | OLS | IV | OLS | IV |
| Change in Chinese imports | −1.068** | −3.266*** | −1.119*** | −2.746*** | −0.376** | −2.041** |
| $\Delta \mathrm{IMP}_{jk}^{CH}$ | (0.453) | (1.148) | (0.227) | (0.735) | (0.168) | (0.930) |
| Change in Chinese imports * technology at $t-5$ | 3.670* | 3.256 | 1.341** | 3.481** | 0.110 | 1.058 |
| $\Delta \mathrm{IMP}_{jk}^{CH} * \mathrm{TECH}_{t-5}$ | (2.162) | (4.609) | (0.509) | (1.584) | (0.441) | (0.763) |
| Technology at $t-5$ | 0.445*** | 0.453*** | 0.239*** | 0.189*** | 0.113*** | 0.076** |
| $\mathrm{TECH}_{t-5}$ | (0.120) | (0.152) | (0.027) | (0.031) | (0.019) | (0.031) |
| First stage $F$-Stat ($\Delta \mathrm{IMP}_{jk}^{CH}$) | | 11.7 | | 11.6 | | 10.45 |
| First stage $F$-Stat ($\Delta \mathrm{IMP}_{jk}^{CH} * \mathrm{TECH}_{t-5}$) | | 2.8 | | 11.7 | | 7.66 |
| Number of units | 1,388 | 1,388 | 2,891 | 2,891 | 12,247 | 12,247 |
| Number of country by industry clusters | 140 | 140 | 83 | 83 | 177 | 177 |
| Observations | 2,377 | 2,377 | 2,891 | 2,891 | 20,625 | 20,625 |
| **Panel B: Survival** | | | | | | |
| Dependent variable: SURVIVAL | (1) | (2) | (3) | (4) | (5) | (6) |
| Method | OLS | IV | OLS | IV | OLS | IV |
| Sample | PATENTS | PATENTS | IT | IT | TFP | TFP |
| Change in Chinese imports | −0.182 | −0.271 | −0.483** | −1.162*** | −0.220*** | −0.308* |
| $\Delta \mathrm{IMP}_{jk}^{CH}$ | (0.171) | (0.247) | (0.200) | (0.383) | (0.083) | (0.142) |
| Change in Chinese imports * Technology at $t-5$ | 0.432** | 0.617* | 0.175 | 0.163 | 0.209* | 0.243 |
| | (0.211) | (0.347) | (0.359) | (0.637) | (0.110) | (0.159) |
| Technology at $t-5$ | −0.024 | −0.030 | 0.002 | −0.015 | −0.017 | −0.018 |
| $\mathrm{TECH}_{t-5}$ | (0.027) | (0.032) | (0.014) | (0.018) | (0.009) | (0.012) |
| First stage $F$-Stat ($\Delta \mathrm{IMP}_{jk}^{CH}$) | | 18.4 | | 13.2 | | 8.68 |
| First stage $F$-Stat ($\Delta \mathrm{IMP}_{jk}^{CH} * \mathrm{TECH}_{t-5}$) | | 17.9 | | 10.7 | | 7.04 |
| Number of industry clusters | 113 | 113 | 84 | 84 | 202 | 202 |
| Observations (and number of units) | 1,624 | 1,624 | 5,980 | 5,980 | 11,794 | 11,794 |

*Notes*: *** denotes 1% significance; ** denotes 5% significance; and * denotes 10% significance. Estimation by OLS in odd numbered columns and IV in even numbered columns. The instrument is "Quota removal" is based on EU SIGL data and defined as the (value weighted) proportion of HS6 products in the four-digit industry that were covered by a quota restriction on China in 2000 (prior to China's WTO accession) that were planned to be removed by 2005 (see online Appendix C for details). We use two instruments for the two endogenous variables in the IV columns, QUOTA and QUOTA * $\mathrm{TECH}_{t-5}$ (the $F$-statistics in this case is the joint test of both instruments). $\Delta \mathrm{IMP}^{CH}$is the five-year difference in Chinese imports as a fraction of total imports in a four-digit industry by country pair. In Columns (1) and (2) TECH is ln[(1+ the firm's patent stock)/employment]; in Columns (3) and (4) TECH is computers per employee and in Columns (5) and (6) TECH is TFP. Twelve countries in all columns except Columns (5) and (6) which is for four countries. Sample is firms in the clothing and textile industry. Standard errors for all regressions are clustered by four-digit industry in parentheses (the quota IV does not vary across within industry across countries like the Chinese import share, which is why we take the more conservative approach compared with Table 1). In Panel A, the dependent variable is the five-year difference of ln(employment). The sample period is 2005–1996 for patents, 2007–2000 for IT, and 2005–1995 for TFP. In Panel B, the sample period is the 2000–2005 cross-section. The dependent variable is SURVIVAL which refers to whether an establishment in Columns (3) or (4) or firm (in all other columns) that was alive in 2000 was still alive in 2005. Specifically, we classify an establishment as having exited if it drops out of the panel and does not appear for four successive years in Columns (3) and (4). In the other columns SURVIVAL it is based on Amadeus company status where exit is defined on the basis of whether a firm that was active in 2000 is recorded as either "bankrupt", "liquidated", or "dormant" in the Company Status variable provided by BVD in 2005 and beyond.

## 5.6 Table 5: Between-firm IV results using quota removal

**Read:**

- Table 5 repeats the reallocation analysis in the textiles / clothing subsample using the quota IV.
- In employment regressions, the interaction between Chinese imports and lagged technology remains positive. The IT interaction is especially strong under IV (3.481 versus 1.341 in OLS).
- In survival regressions, the interaction terms are mostly positive but estimated imprecisely in IV.
- The broad message survives: the reallocation story is not just an OLS artifact, even though the smaller IV sample makes these estimates noisier.

## 5.7 Table 6: Magnitudes

**Read:**

- This table translates coefficients into aggregate importance.
- For patents per worker, China explains 13.9% of overall growth from 2000 to 2007: 5.1% through within-firm upgrading, 6.7% through between-firm reallocation, and 2.1% through exit.
- For IT per worker, the corresponding baseline total is 13.5%; for TFP, also 13.5%.
- Once the paper includes offshoring, the implied contribution rises to 31.8% for IT and 33.7% for TFP.
- The paper's headline "about 14%" number comes directly from this table.

TABLE 6
*Magnitudes*

| Panel A: Increase in patents per employee attributable to Chinese imports | | | | |
|---|---|---|---|---|
| Period | Within | Between | Exit | Total |
| Product market | 5.1 | 6.7 | 2.1 | 13.9 |
| Product market + offshoring | 5.8 | 8.6 | 2.7 | 17.1 |
| **Panel B: Increase in IT per employee attributable to Chinese imports** | | | | |
| Period | Within | Between | Exit | Total |
| Product market | 9.8 | 3.0 | 0.6 | 13.5 |
| Product market + offshoring | 23.2 | 5.4 | 3.2 | 31.8 |
| **Panel C: Increase in total factor productivity attributable to Chinese imports** | | | | |
| Period | Within | Between | Exit | Total |
| Product market | 9.9 | 2.4 | 0.2 | 13.5 |
| Product market + offshoring | 25.0 | 7.6 | 1.1 | 33.7 |

*Notes*: All figures are as a % of the total increase over the period 2000–2007. Panel A reports the share of aggregate patents per worker accounted for by China, Panel B the increase in IT per worker and Panel C the increase in total factor productivity. In each panel the first row (Product Market) uses the coefficients from Tables 1 and 4 to impute the within, between, and total impacts of Chinese import competition on technological as discussed in Section 4.5 and detailed in online Appendix D. The second row also includes the effects of offshoring (see sub-section 5.3).

TABLE 7
*Assessing dynamic selection bias in the patents equation*

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Estimator | Five-year long differences | Five-year long differences | Fixed effects Negative Binomial | Fixed effects Negative Binomial |
| Method | Baseline | Worst-case Lower Bound | Baseline | Worst-case Lower Bound |
| Change in Chinese imports | 0.321*** | 0.271*** | | |
| $\Delta\left(M_{jk}^{\text{China}}/M_{jk}^{\text{World}}\right)$ | (0.102) | (0.098) | | |
| Level of Chinese imports | | | 0.398*** | 0.395*** |
| $\left(M_{jk}^{\text{China}}/M_{jk}^{\text{World}}\right)$ | | | (0.168) | (0.166) |
| Number of clusters | 1,578 | 1,662 | 1,578 | 1,620 |
| Number of firms | 8,480 | 8,732 | 8,480 | 8,732 |
| Number of observations | 30,277 | 31,272 | 74,038 | 74,486 |

*Notes*: *** denotes 1% significance; ** denotes 5% significance; and * denotes 10% significance. Dependent variable is ln(PATENTS) in Columns (1) and (2), and the count of patents in the Negative Binomial specifications in Columns (3) and (4). The sample period is 1996–2005 for all columns. Estimation in Columns (1) and (2) is by OLS in long-differences and by Negative Binomial count data model with fixed effects using the Blundell *et al.* (1999) technique in Columns (3) and (4). Standard errors (clustered by country by four-digit industry pair) in parentheses. "Worst-case lower bounds" impute a value of zero to all observations through 2005 where a firm dies (death is defined as in Table 5 Panel B). There are more observations for the Negative Binomial than five-year long differences as we are using observations with less than five continuous years. All columns include a full set of country by year dummies. Twelve countries included in all samples.

## 5.8 Table 7: Dynamic selection bias in the patents equation

**Read:**

- A natural concern is that only firms with improving technology survive, which could mechanically inflate within-firm patent results.
- The authors construct worst-case lower bounds by assigning zero patents after exit. The OLS coefficient only falls from 0.321 to 0.271.
- A fixed-effects Negative Binomial approach gives 0.398 in the baseline and 0.395 in the worst-case lower bound.
- This is powerful reassurance that the paper's patenting result is not mainly driven by dynamic selection.

## 5.9 Table 8: Initial-conditions IV

TABLE 8
*Using "initial conditions" as an instrumental variable*

| Panel A: Within firm technology equations | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| Dependent variable | $\Delta$ln(PATENTS) | $\Delta$ln(PATENTS) | $\Delta$ln(IT/$N$) | $\Delta$ln(IT/$N$) | $\Delta$TFP | $\Delta$TFP |
| Method | OLS | IV | OLS | IV | OLS | IV |
| Change in Chinese imports | 0.321*** | 0.494** | 0.361*** | 0.593*** | 0.257*** | 0.507* |
| | (0.117) | (0.224) | (0.076) | (0.252) | (0.087) | (0.283) |
| Initial condition IV | | | | | | |
| First-stage $F$-statistic | | 95.1 | | 38.7 | | 14.5 |
| Sample period | 2005–1996 | 2005–1996 | 2007–2000 | 2007–2000 | 2005–1996 | 2005–1996 |
| Number of units | 8,480 | 8,480 | 22,957 | 22,957 | 89,369 | 89,369 |
| Number of industry clusters | 304 | 304 | 371 | 371 | 354 | 354 |
| Observations | 30,277 | 30,277 | 37,500 | 37,500 | 292,167 | 292,167 |

| Panel B: Employment | | | | | | |
|---|---|---|---|---|---|---|
| Dependent variable: Employment growth | (1) | (2) | (3) | (4) | (5) | (6) |
| Technology variable (TECH) | PATENTS | PATENTS | IT | IT | TFP | TFP |
| Method | OLS | IV | OLS | IV | OLS | IV |
| Change in Chinese imports | −0.434*** | −0.734*** | −0.379*** | −1.070*** | −0.377*** | −2.039*** |
| $\Delta IMP_{jk}^{CH}$ | (0.137) | (0.313) | (0.130) | (0.258) | (0.108) | (0.761) |
| Change in Chinese imports * technology at $t-5$ | 1.434** | 0.876 | 0.385** | 1.473*** | 0.795** | 2.443*** |
| $\Delta IMP_{jk}^{CH} * TECH_{t-5}$ | (0.560) | (1.634) | (0.180) | (0.587) | (0.347) | (0.732) |
| Technology at $t-5$ | 0.348*** | 0.365*** | 0.230*** | 0.199*** | 0.136*** | 0.103*** |
| $TECH_{t-5}$ | (0.049) | (0.071) | (0.01) | (0.020) | (0.013) | (0.021) |
| First stage for ($\Delta IMP_{jk}^{CH}$), $F$-stat | | 39.3 | | 22.6 | | 15.6 |
| First stage for ($\Delta IMP_{jk}^{CH} * TECH_{t-5}$), $F$-Stat | | 31.8 | | 24.2 | | 13.3 |
| Number of units | 6,335 | 6,335 | 22,957 | 22,957 | 89,369 | 89,369 |
| Industry clusters | 300 | 300 | 371 | 371 | 354 | 354 |
| Observations | 19,844 | 19,844 | 37,500 | 37,500 | 292,167 | 292,167 |

(*continued*)

TABLE 8
*Continued*

| Panel C: Survival | | | | | | |
|---|---|---|---|---|---|---|
| Dependent variable: Survival | (1) | (2) | (3) | (4) | (5) | (6) |
| Technology variable (TECH) | PATENTS | PATENTS | IT | IT | TFP | TFP |
| Method | OLS | IV | OLS | IV | OLS | IV |
| Change in Chinese imports | −0.078 | −0.428 | −0.182** | −0.797*** | −0.208*** | −0.955*** |
| $\Delta IMP_{jk}^{CH}$ | (0.051) | (0.454) | (0.077) | (0.275) | (0.067) | (0.317) |
| Change in Chinese imports * technology at $t-5$ | 0.260** | 0.793 | 0.137 | 0.490 | 0.110** | 0.314** |
| $\Delta IMP_{jk}^{CH} * TECH_{t-5}$ | (0.114) | (0.494) | (0.117) | (0.471) | (0.055) | (0.159) |
| Technology at $t-5$ | −0.005 | −0.022 | −0.002 | −0.014 | −0.004 | −0.010 |
| $TECH_{t-5}$ | (0.009) | (0.016) | (0.007) | (0.014) | (0.003) | (0.006) |
| First stage for ($\Delta IMP_{jk}^{CH}$), $F$-Statistic | | 12.52 | | 7.86 | | 2.94 |
| First stage for ($\Delta IMP_{jk}^{CH} * TECH_{t-5}$), $F$-Stat | | 15.56 | | 7.73 | | 2.97 |
| Industry clusters | 328 | 328 | 372 | 372 | 379 | 379 |
| Observations (and number of units) | 7,928 | 7,928 | 28,624 | 28,624 | 60,883 | 60,883 |

*Notes*: *** denotes 1% significance; ** denotes 5% significance; and * denotes 10% significance. In Panel A we use the same specifications as Table 1 Panel A from Columns (1), (3), and (5) but estimate by instrumental variables (IV) in the even numbered columns. Similarly in Panels B and C we use Table 4 Panel A and B, respectively (columns (2), (4), and (6)), for the odd numbered columns in Table 8. IV equivalents are in even numbered columns. The Initial Conditions IV is the share of Chinese imports (in all imports) in the four-digit industry across the whole of the Europe and the US (six years earlier) interacted with the aggregate growth in Chinese imports in Europe. In Panels B and C we have two instruments: the linear initial conditions and the initial conditions interacted with $TECH_{t-5}$ (the $F$-statistics in this case is the joint test of both instruments). The number of units is the number of firms in all columns except the IT specification where it is the number of plants. All columns include country by year effects. Standard errors for all regressions are clustered by four-digit industry in parentheses.

**Read:**

- This table shows the alternative IV strategy based on lagged Chinese comparative advantage.

- Panel A again finds larger IV than OLS coefficients: 0.494 versus 0.321 for patents, 0.593 versus 0.361 for IT, and 0.507 versus 0.257 for TFP, with strong first stages.
- Panel B shows employment results with the same qualitative pattern as before: Chinese imports reduce jobs and the interaction with lagged technology remains positive.
- Panel C shows survival effects that are directionally similar, though first stages are weaker and the estimates are less precise.
- This table matters because it extends the IV evidence beyond textiles and clothing.

# 6 Short summary

- **Conclusion.** The paper finds that Chinese import competition induced technical change in Europe, both by making surviving firms innovate and adopt IT faster and by reallocating activity toward more advanced firms.
- **Intuition.** Low-wage import competition makes old, low-tech production less attractive, so firms either move up the technology ladder or shrink and exit.
- **Empirical lesson.** To understand trade and productivity, it is not enough to study aggregate productivity or firm exit alone; one needs direct within-firm technology measures like patents and IT.

# 7 Conclusion

## 7.1 Core conclusion (what the paper establishes)

- **Main finding.** Chinese import competition raised patenting, IT intensity, and TFP within incumbent European firms.
- **Reallocation result.** Chinese imports also reduced employment and survival among firms with weaker initial technology, generating a between-firm upgrading effect.
- **Magnitude.** Baseline estimates imply that China accounted for about 14% of European technology upgrading from 2000 to 2007, with even larger effects when offshoring or IV estimates are considered.
- **Credibility.** The quota-based IV, the trend-adjusted reduced forms, the initial-conditions IV, and the dynamic-selection-bias bounds all point in the same direction.
- **What the paper rules out.** The story is not just defensive patenting, not just product switching, and not just a generic import effect from rich countries.

## 7.2 Economic intuition

- Trade with China compresses the rents from producing mature, low-tech goods in Europe.
- Firms that can respond by innovating, adopting computers, reorganizing production, or offshoring low-value tasks survive and become more technologically advanced.

- Firms that start with weak technology are less able to adapt, so they lose jobs or exit. This reallocation further raises average technology in exposed sectors.
- In that sense, the paper combines a Schumpeterian within-firm response with a Melitz-style between-firm selection response.

## 7.3 Takeaway

This paper's key contribution is to show that the China shock was not only a labour-market or trade event. It was also a technology event. The rise of Chinese imports pushed European firms to innovate more, adopt more IT, and move resources away from technologically backward producers. Any welfare analysis of trade with China therefore has to weigh not only job losses and distributional costs, but also the dynamic gains from faster technical change.